

Videxa: Intelligent Video Knowledge Retrieval System Using Retrieval-Augmented Generation (RAG)

Background

With the rapid growth of online learning platforms, recorded lectures, tutorials, meetings, interviews, webinars, and sports broadcasts, users consume a vast amount of video content daily. However, extracting specific information from long videos remains a significant challenge. Users often spend considerable time searching through entire recordings to locate important concepts, explanations, decisions, or events.

Traditional video platforms provide basic playback controls and limited keyword-based searching, which do not fully understand the context of the video content. As a result, finding precise information, revising concepts, generating notes, or obtaining quick summaries becomes time-consuming and inefficient.

Problem Statement

There is a need for an intelligent system that can transform video content into a searchable knowledge source. Users should be able to upload a video and interact with its content through natural language queries instead of manually watching or rewatching the entire recording.

The system must automatically extract spoken content from videos, understand the context, store the information efficiently, and retrieve relevant knowledge when users ask questions. It should provide accurate answers, summaries, notes, quizzes, concept explanations, and timestamp references based solely on the uploaded video's content.

The challenge is to develop a scalable Retrieval-Augmented Generation (RAG) framework that combines speech recognition, semantic search, vector databases, and large language models to enable intelligent interaction with video content.

Proposed Solution

Videxa is an AI-powered Video RAG platform that converts uploaded videos into interactive knowledge bases. The system extracts audio from videos, generates transcripts, creates embeddings, stores them in a vector database, and uses Retrieval-Augmented Generation to answer user queries.

Users can ask questions, generate summaries, create notes, build quizzes, and retrieve timestamp-specific explanations directly from the uploaded video.

Objectives

1. Develop a system for uploading and processing video files.
2. Extract audio and generate accurate transcripts.
3. Implement a RAG pipeline for semantic retrieval and question answering.
4. Generate concise video summaries.
5. Create structured study notes automatically.
6. Provide timestamp-based answer retrieval.

7. Generate quizzes and MCQs from video content.
8. Build concept maps from extracted topics.
9. Support revision and doubt-solving modes.
10. Improve knowledge accessibility from long-form video content.

Expected Outcomes

- Faster access to information within videos.
- Improved learning and revision efficiency.
- Reduced need to rewatch lengthy recordings.
- Intelligent video-based question answering.
- Automatic note and quiz generation.
- Enhanced educational and professional productivity.

Scope of Application

Educational lectures, online courses, technical tutorials, meeting recordings, interview recordings, webinars, sports commentary analysis, corporate training videos, and research presentations.

Innovation

Unlike traditional video players, Videxa transforms videos into interactive knowledge systems where users can search, question, summarize, revise, and learn directly from video content using Retrieval-Augmented Generation and AI-powered semantic understanding.